Build the one strand

68 tiles, and every piece of one tile on a single strand

The tile is the straight-edged form of the Spectre, an aperiodic monotile discovered in 2023 by David Smith, Joseph Samuel Myers, Craig S. Kaplan and Chaim Goodman-Strauss.

The Spectre Knot is a Spectre tile with 14 pieces of ribbon drawn on it. When tiled in the right way (the nest tiling below), they join into one endless strand: an infinite single strand Celtic knot on an aperiodic monotile. Pictures and the cut-out tile: cortado-j.github.io/SPECTRE_KNOT.

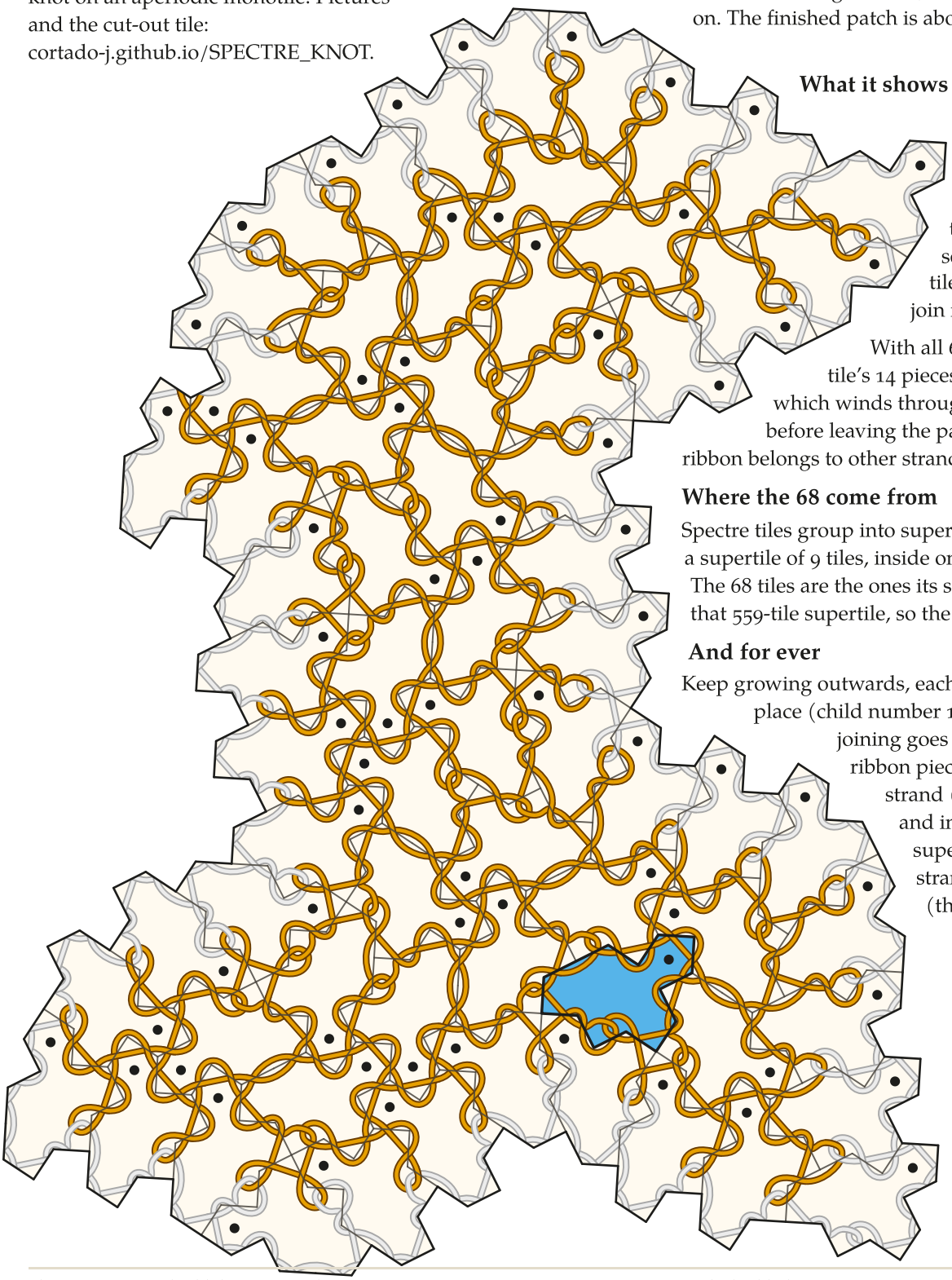
How to build it

1. Print pages 2 to 13 at actual size (a tile edge is 25.7 mm). Cut out the 72 tiles on their black lines: 68, and 4 spare.

2. Put down the blue tile first: the starting tile.

3. Lay the others edge to edge round it, as in the plan. Each tile's QR code sits in its head, the lobe with four sides; on the plan a dot • marks each head. Turn each tile so its QR code is where the plan shows a dot. Never turn a tile over.

4. Where two edges touch, the ribbons meet and carry on. The finished patch is about 67 × 90 cm.



What it shows

Every tile carries the same drawing: 14 pieces of ribbon and one crossing. On the starting tile alone, they are 14 separate bits of ribbon. As tiles go down round it they join into longer strands.

With all 68 tiles down, the starting tile's 14 pieces lie on **one strand**, which winds through every one of the 68 tiles before leaving the patch at both ends. The grey ribbon belongs to other strands.

Where the 68 come from

Spectre tiles group into supertiles: the starting tile sits in a supertile of 9 tiles, inside one of 71, inside one of 559. The 68 tiles are the ones its strand passes through in that 559-tile supertile, so the plan is a piece of it.

And for ever

Keep growing outwards, each block sitting in the same place (child number 1) inside the next, and the joining goes on. In this *nest tiling* all the ribbon pieces join into one endless strand (the Ariadne Theorem); and in every spectre tiling whose supertiles fill the plane, no strand ever closes into a loop (the Daedalus Theorem).

The Spectre Knot

A single-strand Celtic knot
on an aperiodic monotile

cortado-j.github.io/SPECTRE_KNOT



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cut on the black lines


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